

NIKHIL YEMINEDI

Atlanta, GA · (404) 203-8442 · nikhilgoudyeminedi@gmail.com · [LinkedIn](#) · [GitHub](#)

PROFESSIONAL SUMMARY

Applied NLP and ML Research Engineer with 3.5+ years of production experience building large language model fine-tuning pipelines, transformer-based NLP systems, and knowledge graph architectures across fintech, legal-tech, and academic research. From engineering fault-tolerant financial pipelines at Visa to publishing research that opens the black box of AI had deep hands-on expertise in BERT/RoBERTa fine-tuning, mechanistic interpretability, entity resolution, RAG architecture, and scalable NLP backends on AWS.

TECHNICAL SKILLS

LLM & NLP: BERT, RoBERTa, Hugging Face Transformers, Fine-Tuning, Transfer Learning, Mechanistic Interpretability, Attention Analysis, NER, Entity Resolution, spaCy, PyTorch, TensorFlow, MLflow, scikit-learn

NLP Research Methods: Co-occurrence metrics (LLR, PMI, DSC), t-SNE latent space clustering, Welch T-test, Causal ablation, Induction head mapping, Knowledge graph generation, Neuro-symbolic frameworks

GenAI & Retrieval: RAG Architecture, Prompt Engineering, Embedding-based retrieval, Context management, LLM integration (Claude, BERT), Vector search, HuggingFace

ML Frameworks & Research: CNNs, RNNs, LSTMs, GBDT, Federated Learning, Adversarial robustness, Graph Neural Networks (GNN), Deep Learning, Model Evaluation, Distributed Training

Data & Engineering: Python (advanced), SQL, PySpark, Apache Spark, Airflow DAGs, ETL/ELT Pipelines, PostgreSQL, MongoDB, Snowflake, Databricks, Flask, Django, REST APIs, Docker

Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, IAM, VPC), Microsoft Azure, Git, GitHub Actions, Jenkins, CI/CD, Linux/Unix, Power BI, Matplotlib

Certifications: AWS Certified Cloud Practitioner · Moving Data into Hadoop – IBM · Python & Flask – Udemy

PROFESSIONAL EXPERIENCE

NLP / ML Engineer · [SCALES](#) · Atlanta, GA

May 2025 – Present

- Fine-tuned **RoBERTa-based Entity Resolution Engine (Ditto framework)** on 1,000+ unstructured legal documents from 70+ source types; built zero-labeled-data training pipeline generating positive pairs (abbreviations, typos, nicknames) and hard negatives achieved 100% F1 on entity matching and 99.9% record-linkage accuracy.
- Built **spaCy NER microservices with caching and concurrency controls** processing **15M+ U.S. Federal Court** (PACER) records; improved named-entity extraction precision by 24% and achieved 98% entity-resolution accuracy across heterogeneous document formats.
- Constructed a **multi-entity knowledge graph** resolving 1,600+ entity mentions into 1,190 unique entities (4 node types, 4 relationship types) via Union-Find clustering; engineered multi-key blocking strategies reducing pairwise comparison space by 95% while maintaining full recall.
- Built a production **RAG-based legal AI assistant** integrating LLM architecture (Claude, BERT) with structured legal datasets; implemented prompt engineering, embedding-based retrieval, and context management for real-time intelligent legal query resolution with 99.9% uptime on AWS EC2.
- Delivered **Power BI dashboards and SQL-backed analytical reports** achieving 100% stakeholder workflow adoption; led technical workstream for civic-tech hackathon architecting eviction-prediction pipelines with Census API integration and ML classification.

AI Research Engineer · [DICE Lab, Georgia State University](#) · Atlanta, GA

Jan 2025 – Dec 2025

- Fine-tuned **BERT classifier on 163K+ annotated social media posts and news articles** for disaster event detection; achieved 99% classification accuracy — 43% improvement over baseline — through hyperparameter optimization and distributed Spark compute. Published at NLPiR 2026 (Springer).
- Designed a **mechanistic interpretability system intercepting BERT forward passes** in real time; single-pass O(1)-memory architecture using tensor masking extracts [CLS] attention arrays across all 12 Transformer layers and 144 attention heads — second paper under review, targeting ACL/EMNLP/NAACL 2026.
- Implemented **corpus-level interpretability probes**: t-SNE latent space clustering, Welch's T-test statistical lexicon validation ($p < 0.0001$), layer-wise attention evolution analysis, and induction head specialization mapping; causal ablation proved masking lexicon tokens collapses model confidence from 99.5% to 2.6%.
- Built **statistical validation framework** using co-occurrence metrics (LLR, PMI, DSC) for knowledge graph generation and model bias evaluation; developed full-stack disaster informatics application (Flask + React + PostgreSQL) integrating fine-tuned LLM classifiers with **US NOAA meteorological datasets** as ground-truth. Reduced manual analysis effort by 70%.

Data & ML Engineer · [Visa Inc. \(via TCS\)](#) · India

Jul 2022 – Dec 2024

- Applied **ML for anomaly detection, credit risk forecasting, and predictive analytics** on millions of daily financial transactions for Visa's Global Federated Data Mart (GFDM); achieved 28% faster processing, 99.7% pipeline reliability, and contributed to \$10M+ in cost savings on the TUSKER project.
- Engineered and automated **30+ production ETL/ELT pipelines** using Python, Apache Spark, Hadoop, Databricks, and SQL; improved data integration speed by 42%, reduced manual effort by 60%, and delivered Power BI dashboards backed by Snowflake and Oracle for executive real-time analytics.
- Migrated large-scale datasets across distributed Hadoop clusters using **DistCP and Apache Sqoop**; implemented Jenkins CI/CD pipelines with Docker achieving 97% deployment success; supported enterprise credit risk analysis and financial data standards.

Full-Stack Developer · *Steed 26 Solutions Pvt. Ltd.* · India

Mar 2021 – Jun 2022

- Delivered Hospital Records Management System (HRMS) serving **1,000+ concurrent users** using Java, JavaScript, SQL, HTML5/CSS3, and REST APIs; improved data retrieval efficiency by 55% and reduced page load time by 30% through backend optimization and relational database schema design.
- Maintained **95%+ code quality** through Agile Scrum, Git version control, and QA automation; integrated RESTful service layers and structured database pipelines supporting scalable healthcare data workflows.

RESEARCH PUBLICATIONS

[Published] Shrestha, A., Yeminedi, N., Kubek, M. M., & Mikler, A. R. (2026). Interpretable Adverse Event Detection Using a Neuro-Symbolic and Graph-Based Approach. Proceedings of NLPiR 2026. Springer.

[Under Review] Yeminedi, N., Shrestha, A., Kubek, M. M., & Mikler, A. R. (2026). Towards Attention-Based Mechanistic Interpretability in Adverse Event Detection. DICE Lab, GSU. Target venue: ACL / EMNLP / NAACL 2026.

KEY PROJECTS

LLM Fine-Tuning Pipeline for Disaster Event Classification | BERT · RoBERTa · PyTorch · HuggingFace · Apache Spark · MLflow Jan 2025 – Aug 2025

- Production-grade LLM fine-tuning pipeline on 163K+ labeled disaster tweets; applied MLflow for experiment tracking and model versioning; co-occurrence metrics (LLR, PMI, DSC) for bias evaluation and interpretability — **99% accuracy, 43% over baseline**. Core of published NLPiR 2026 research.

Heterogeneous Graph Neural Network for Entity Resolution | Python · RoBERTa · spaCy · NetworkX · Docker May 2025 – Present

- End-to-end entity resolution system using fine-tuned RoBERTa and graph-neighborhood similarity scoring; **100% F1 on entity matching** across 1,000+ documents from 70+ sources; multi-entity knowledge graph resolving 1,600+ mentions into 1,190 unique entities; multi-key blocking reduced pairwise space by 95%. Deployed via Docker and REST APIs.

Attention-Based Mechanistic Interpretability Framework | BERT · PyTorch · Tensor Masking

Jan 2025 – Present

- Single-pass O(1)-memory architecture extracting [CLS] attention arrays across all 12 Transformer layers and 144 attention heads; implemented t-SNE clustering, Welch T-test validation ($p < 0.0001$), induction head mapping, and causal ablation — confidence collapse from **99.5% → 2.6%** on lexicon masking. Under review at ACL/EMNLP 2026.

AI-Powered Eviction Data Hub — Intelligent Extraction, Classification & Geocoding | Python · PySpark · Airflow · Census API · AWS S3 · PostgreSQL · Power BI May 2025 – Aug 2025

- End-to-end intelligent pipeline processing **200,000+ eviction court cases** with NLP-based entity extraction, fuzzy matching, ML classification, automated geocoding, and Airflow DAG orchestration; Census API enrichment for demographic analysis; feeds downstream housing equity research models.

Financial Analytics & Business Intelligence Platform | Python · Firebase · Power BI · Snowflake · SQL ·

Jan 2025 – Apr 2025

- End-to-end financial data platform integrating Python ETL pipelines, Firebase data warehousing, **predictive analytics and forecasting models**; delivered interactive Power BI dashboards with real-time API integration enabling executive BI across multi-source datasets.

EDUCATION

M.S. Computer Science · Georgia State University · Atlanta, GA

Jan 2025 – Aug 2026 GPA: 4.0/4.0